

# 2017 RAA

Name ANSWERS

## Pre-Calculus

School \_\_\_\_\_

(Use full school name – no abbreviations)

\_\_\_\_\_ Correct X 2 pts. ea. =

**Note:** All answers must be written legibly in simplest form, according to the specifications stated in the Contest Manual. Exact answers are to be given unless otherwise specified in the question. No units of measurement are required.

1.  $-\frac{3}{2}$  OR  $-\frac{3}{2}$  OR  $-1.5$  OR  $-1\frac{1}{2}$

2.  $-\frac{1}{3}\pi$  OR  $-\frac{\pi}{3}$  OR  $-\frac{\pi}{3}$  (Must be this exact answer.)

3.  $-1$

4.  $133.962$  (Must be this exact decimal.)

5.  $120$

6.  $6.325$  (Must be this exact decimal.)

7.  $15$

8.  $800$  ("feet" or "ft." optional.)

9.  $13$

10.  $\text{ellipse}$  (Must be the whole word, caps optional.)

11.  $\left(\frac{\sqrt{6}}{2}, \frac{\sqrt{2}}{2}\right)$  (Must be this ordered pair with exact values.)

12.  $1.738$  (Must be this decimal.)

13.  $\left(4, \frac{4}{3}\pi\right)$  OR  $\left(-4, \frac{\pi}{3}\right)$  (Either exact coordinate pair accepted; only one.) answer needed)

14.  $294$

15.  $55.25$  (Must be this decimal, "inches" or "in." optional.)

16.  $3$

17.  $39.482$  (Must be this decimal, "degrees" or "°" opt.)

18.  $3$

19.  $\frac{9}{2}$  (Must be this reduced improper fraction.)

20.  $y = 4$  (Must be this equation.)